DEPARTMENT OF THE AIR FORCE
DEAN OF FACULTY
DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING
UNITED STATES AIR FORCE ACADEMY

1. COURSE IDENTIFICATION

Course Number:	ECE 485
Course Title:	Advanced Computer Architecture
Semester Hours:	3.0
Term:	Spring 2027
Department:	Department of Electrical and Computer Engineering
Prerequisites:	ECE 382 and either Math 220, 300, 356, or 377
Co-requisites:	None

2. COURSE DESCRIPTION

This final course in the Computer Systems Area of Study quantitatively examines trade-offs in the design of high-performance computer systems. Topics include price/performance, instruction sets, hardwired control versus microprogramming, memory hierarchy, cache memory, virtual memory, pipelining, reduced instruction set computers (RISC), input/output, and parallel processing. The Project will examine state-of-the-art processors and computers. Lab. Final Exam. - 3 credits spring.

SECTIONS 3-9 Not for PA Release

3. COURSE DIRECTOR / INSTRUCTOR INFORMATION

3a. Course Director

Name / Rank:	Dr. George York
Office:	Fairchild, 2E46
Phone:	719-333-3190
Email:	george.york@afacademy.af.edu
Office Hours:	Monday-Friday 0800-1600 or by appointment

3b. Instructors

Instructor:	Dr. George York, Fairchild 2E46, george.york@afacademy.af.edu

4. STUDENT LEARNING OUTCOMES

4a. Course Objectives

Cadets shall be able to:

1. Explain fundamental concepts and analyze design trade-offs associated with high performance computers and computer networks.
2. Quantitatively analyze and compare the performance of various computer architectures.

4b. Accreditation Outcomes: Not applicable.

5. REQUIRED MATERIALS

#	Title / Description	Author(s)	Edition / Notes
1	***Computer Architecture: A Quantitative Approach*** **.**	Hennessy & Patterson	6th Edition

6. GRADING

6a. Assignments/Course Assessments

Component	Prog	Final
HW/Quizzes/CPH/Lab	*50%*	*15%*
Skills Review	*10%*	*0%*
GRs(2)	*40%*	*40%*
CPX(4)	*0%*	*20%*
Final Exam	*0%*	*25%*
TOTAL	***100%***	**100%**

6b. Grading Scale

A	A-	B+	B	B-	C+	C	C-	D	F
100%-93%	*92% - 90%*	*89% - 87%*	*86% - 83%*	*82% - 80%*	*79% - 77%*	*76% - 73%*	*72% - 70%*	*69% - 60%*	*59% - 0%*

6c. Assignment Description

Quizzes	There are typically 3 quizzes. Quizzes may or may not be announced.

GRs	There are two GRs. Primary sources for questions include the textbook, information presented in class, handouts, and homework problems. All exams are closed textbook and notes. Students are allowed an equation sheet, for GR1, the front side of a 8x11.5 sheet of paper, and for GR2, both sides of a 8x11.5 sheet of paper. Only equations may be on the equation sheet. Worked out problems and vocabulary definitions are not allowed. GRs are individual effort. You must notify your instructor one lesson in advance to schedule a make-up if you anticipate missing any scheduled graded evaluation. Schedule makeup exams before missing the exam!
Skills Review	There is a skill review, primarily of your ECE 281 knowledge, due lesson 4. This will be averaged with your quiz grade
RISC/CISC Debate	Around lesson 8, you will be assigned a team to compete in the RISC/CISC Debate. This grade will be averaged with your quiz grade.
CPH	Computer Homeworks are due at 0800 the day of your class. These are a combination of preflights over the reading and homework, and a usually online via blackboard or gradescope.
RISC-V Exercise	this lab will be due around lesson 13 to prepare for the final project
CPX	There will be 4 Computer Exercises (CPX), implementing a pipelined processor in VHDL. Students must complete all 4 stages, meeting at least ”basic functionality.” Failure to meet ”basic functionality” in all 4 stages will result in a ”controllable incomplete” grade, until all stages are complete.
Final Exam	The Final Exam will cover all 41 lessons of the course.

7. COURSE POLICIES

Attendance	1. Be in class on time or notify your instructor **(via e-mail with copy to Sq Commander/AMT)** of your whereabouts/reason for absence (medical, TDY, etc.). Provide SCA# if appropriate but be sure to indicate **where** you will be. Instructors cannot excuse cadets from class and must be accountable for all absences. 2. If you are approved for Sq Commander bedrest, notify instructor immediately and copy your Sq Commander/AMT on email or Teams message.
Academic Integrity	See the Cadet Wing Honor Code for details. Expect and require the highest standards of military courtesy, professionalism, and academic rigor.

Late Work	CPH's must be turned in on time (else a zero if late). For all other assignments, the late turn-in penalties are: 10% for the first calendar day. 30% for the second calendar day. 60% for the third calendar day. 100% for four or more calendar day (at this point you earn a 0). If notified in advance, an instructor can make individual exceptions to this policy or possibly grant extensions. Regarding extensions on the CPX: The schedule for the due dates for the CPX are published lesson 1. Do not expect any extensions. For example, if going on a trip the week a stage is due, you are expected to turn it in BEFORE you leave… schedule your time appropriately.
EI	The time to seek help is when you have difficulty understanding the material and you have attempted to work the assigned problems. I will be glad to schedule EI at mutually convenient times but be sure to call or email if you must cancel. As always, if you have a burning question, or find yourself in a general state of confusion, feel free to try and find me for an answer.

Collaboration Policy	For CPHs you are allowed to work with other students, use your textbook and other instructor-provided course materials, but not allowed to use Generative AI like ChatGPT or search engines, like google. For the skills review, you are allowed to work with other students, use your 281 textbook and other instructor-provided course materials. For the RISC versus CISC debate, you are allowed to collaborate with students on your own team, use your textbook, and other instructor-provided course materials. For the RISC-V exercise, the only collaboration allowed with classmates is for general questions, such as how the 9 assembly instructions work and relate to the block diagrams of figures C.18, C.19 and C.25. All the work filling out the spreadsheets and answering the above questions must be your own work. Do not look at each other's spreadsheets or answers to the above questions. For the CPX, the only collaboration allowed with classmates is for general questions, such has how to implement VHDL code, like "how to make a process statement to implement a register", not for specific coding in this assignment, or for general concepts regarding the blocks in the block diagram. All coding must be your own work. Do not look at each other's code. You are not allowed to have another student look at your code to help with debugging, unless given specific authorization from your instructor. Regardless of the assignment, all collaboration must be clearly documented; copying another person's work, with or without documentation, will result in NO academic credit.

Use of AI Tools	For CPHs you are not allowed to use Generative AI like ChatGPT or search engines, like google (GenAI level 0: No use of GenAI) For the skills review, you are allowed to Generative AI like ChatGPT or search engines, like google at GenAI level 1: Organizational / Explanatory use of GenAI. For the RISC versus CISC debate, you are allowed to Generative AI like ChatGPT or search engines, like google at GenAI level 5: Unrestricted, attributed use of GenAI. For the RISC-V exercise, regarding the use of GenAI or search engines is GenAI level 1: Organizational / Explanatory use of GenAI. For example, you can prompt GenAI on how the 9 assembly instructions work and relate to the block diagrams of figures C.18, C.19 and C.25; however, all the work filling out the spreadsheets and answering the above questions must be your own work. Do not ask GenAI to produce specific answers to the spreadsheets or the above questions. For the CPX, the use of generative AI (e.g., ChatGPT) or search engines is "level 2: Use of GenAI for brainstorming or idea generation." For example, you can prompt GenAI on how to implement basic hardware in VHDL, such as "how to you implement a 32-bit register in VHDL?" "How to you implement a multiplexer using the when statement?" or ask basic questions about the RISC-V multicycle architecture, like "why is the ALU controller separate from the Control Unit in RISC-V?"; however, any coding must be your own work. Do not ask GenAI to produce specific code for this assignment, like "Given this VHDL code for the ALU, add the SUBI function." Regardless of the assignment, any use of GenAI or internet searches must be clearly documented, including all prompts used. Failure to follow this policy will result in NO academic credit. One warning about GenAI and RISC-V: besides the common problem of GenAI hallucinations, there are different versions of RISC-V, different from this assignment, so GenAI could lead you astray.

8. Additional Information

Class website: https://usafa-ece.github.io/ece485/intro.html The class website mainly has the syllabus, schedule and some information about the assignments. The handouts referred to in the schedule will be posted on the class website.

Class Team's channel: We will mainly use the Team's channel for messaging and for posting the class lesson slides.

Class Blackboard site: blackboard for ECE 485 202730-G-DFEC-ECE485-2MA: ADVANCED COMPUTER ARCHITECTURE,

This has the assignments like online homework/preflights called CPHs.

CPH1 is due lesson 2

Skills Review is due lesson 4

GradeScope: We will also use GradeScope for some assignments, like the skills review and the CPX.

https://www.gradescope.com/courses/1196106

book EI: To book EI or a meeting with Dr York, I will give out a booking link.

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9. Course Schedule

Lesson	Topic	Pre-Flight	Graded Event
1	Course Introduction	- read 1.1 - 1.6 - see skills review assignment (due lesson 4); on Gradescope - lesson slides on Teams Site	
2	Skill Review Review Day [Single-Cycle RISC-V architecture]	- CPH1 due 0800 - read Microarchitecture.pdf - other readings: RandI.pdf; Registers.pdf; Branching.pdf; - lesson slides on Teams Site	
3	Trends in Computer Technology	- CPH2 due 0800 - read 1.8 - 1.9 - lesson slides on Teams Site	
4	Price vs Performance	- Skills Review due 0800; on Gradescope - read 1.6 - 1.9 - lesson slides on Teams Site	
5	Performance, Metrics, Benchmarks, Amdahl's Law	- CPH4 due 0800 - read 1.10-1.12 - benchmarking articles: PerformanceSingleNumber.pdf; MeasuringPerformance.pdf; BenchmarkDescriptions.pdf - lesson slides on Teams Site	

6	Metrics and Benchmarks	- CPH5 due 0800 - read 1.10-1.12 - read A.1 - A.9 - lesson slides on Teams Site	Quiz today?
7	Instruction Set design, memory addressing, operations, operands	- CPH6 due 0800 - read A.1 - A.9 - CaseforRISC.pdf - lesson slides on Teams Site	
8	CISC vs RISC Debate	- CPH7 due 0800 - read RISCvsCISCdebate.pdf - lesson slides on Teams Site	RISC versus CISC Debate
9	Compiler Technology, RISC-V architecture	- Sorry, no CPH8 this year - read A.8 - A.9 - lesson slides on Teams Site	
10	Pipeline Introduction and Hazards	- CPH9 due 0800 - read C.1 - C.5, C.8 - lesson slides on Teams Site	
11	Data Hazards, Control Hazards, Pipeline Implementation, ILP	- CPH10 due 0800 - read C.3 - C.5, C.8, 3.1 - L10Worksheet.docx - lesson slides on Teams Site	
12	Overcoming Hazards: forwarding and branch fix	- CPH11 due 0800 - read C.3 - C.5, C.8, 3.1 - L11Worksheet.docx - see RISCV Exercise, due L13 taps, on blackboard and website. - lesson slides on Teams Site	
13	RISC-V Exercise, In-class	- CPH12 due 0800 - see RISCV Exercise, due L13 taps, on blackboard and website. - lesson slides on Teams Site	

14	Dependencies: Data, Name, Control	- Sorry, no CPH13 this year - read C.7, 3.4 - PipeliningandSuperscalarExecution.docx - L12Worksheet.docx - lesson slides on Teams Site	
15	Overcoming Hazards - Dynamic Scheduling and Tomasulo's Algorithm	- CPH14 due 0800 - read 3.5 - other readings: MultipleArithmeticUnits.pdf; L14Worksheet.docx; L15Worksheet.docx; TomasuloExample.pdf; AFITTomasulo.ppt - lesson slides on Teams Site	
16	Tomasulo's Algorithm and Dynamic Branch Prediction	- CPH16 due 0800 - Sorry, no CPH15 this year - read 3.3, 3.6 - lesson slides on Teams Site	
17	Branch Target Buffers, Speculation	- CPH17 due 0800 - read 3.7 - 3.9, 3.12-3.14, 5.8 - L17Worksheet.docx - lesson slides on Teams Site	
18	GR1		GR1
19	CPX Stage1	- see Stage 1 Assignment, due taps M21	
20	CPX Stage1		
21	VLIW and Superscalar, SIMD?	- Stage 1 Assignment, due taps M21 - read 4.3 MultiscalarProcessors.pdf	
22	Memory Hierarchy Design: Cache Memory	- CPH20 due 0800 - read 2.1, B.1, B.2 - L20Worksheet.docx - lesson slides on Teams Site	
23	Memory Hierarchy Design: Cache Memory	- CPH21 due 0800 - read 2.1, B.1, B.2 - L21Worksheet.docx - lesson slides on Teams Site	

24	Improving Cache Performance	- CPH22 due 0800 - read 2.1, B.1, B.2, B.3, B.4 - lesson slides on Teams Site	Quiz today?
25	Improving Cache Performance; Main Memory	- CPH25 due 0800 - read 2.2, B.3, B.4 - L22Worksheet.docx - lesson slides on Teams Site	
26	Virtual Memory	- CPH26 due 0800 - read 2.4 - lesson slides on Teams Site	
27	CPX Stage2	- see Stage 2 Assignment, due taps L29	
28	CPX Stage2		
29	Storage Devices: RAID	- CPH27 due 0800 - read D.1 - D.3 - skim D.4, D.7 - see appendD.pdf - lesson slides on Teams Site	
30	I/O Interfacing and Performance, Interconnection Networks	- CPH29 due 0800 - read F.1, F.2 - skim F.5, F.6 - see appendF.pdf - lesson slides on Teams Site	
31	LAN Topologies, Routers/Gateways, Flynn's Taxonomy; Memory Archs	- CPH30 due 0800 - read F.1, F.2 - skim F.5, F.6 - see appendF.pdf - L29Worksheet.docx - lesson slides on Teams Site	
32	Introduction to Multiprocessors	- read page 10; 5.1 - L30Worksheet.docx - lesson slides on Teams Site	
33	Symmetric Shared Memory Architecture	- CPH31 due 0800 - read 5.2 - L31Worksheet.docx - lesson slides on Teams Site	
34	CPX Stage3	- see Stage 3 Assignment, due taps L36	

35	CPX Stage3		
36	CPX Stage3		
37	Distributed Memory Coherence	- CPH32 due 0800 - read 5.4 - skim 5.5, 5.6 - L32Worksheet.docx - lesson slides on Teams Site	
38	CPX Stage4	- see Stage 4 Assignment, due taps L41	
39	GR2		GR2
40	CPX Stage4		
41	CPX Stage4		